

HENGRONGFEI LI



CONTACT

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SKILLS

- Programming & Development: Python, MATLAB, C/C++, Git
- Robotics: Industrial robots, cobots, UAVs, humanoid robots, quadruped robots
- Modeling & Simulation: ROS2 Jazzy/Humble, Simulink
- Computer Vision & AI: OpenCV, YOLO, PyTorch, TensorFlow, Transformers, LLMs
- Data Acquisition & Measurement: Modbus, ultrasonic sensing, acoustic measurement, real-time data logging, InfluxDB
- Visualization: PyQt, Streamlit, Matplotlib

LANGUAGES

- Chinese: native
- English: business fluent
- German: business fluent
- Japanese: basic knowledge



PROFILE

Master's student in Electrical Engineering focused on robotics, automation, and intelligent systems. Experienced in robotic system integration, computer vision, machine learning, and real-time data acquisition across industrial and mobile robotic platforms.



EDUCATION

Technical University of Munich 04.2025 - PRESENT
Master of Science in Electrical Engineering and Information Technology
(Focus: Automation and Robotics)

Technical University of Munich 10.2021 - 04.2025
Bachelor of Science in Electrical Engineering and Information Technology



WORK EXPERIENCE

Working student, TÜV SÜD 03.2025 - 03.2026

- Supported product safety testing according to EN/IEC and UL standards for electrical and electronic systems.
- Analyzed and documented measurement results from laboratory and office-based testing.
- Built an automated measurement & monitoring pipeline (Modbus → InfluxDB, real-time dashboards)

Bachelor Thesis, Fraunhofer IPMS 11.2024 - 04.2025
Thesis title: Ultrasonic-Based Object Recognition for Industrial Applications

- Conducted pulse-echo measurements and applied advanced signal processing
- Developed and trained CNN/LSTM models for distance and angle prediction
- Achieved high accuracy over 95% in object localization, demonstrating applicability for industrial robotics.

Internship, Fraunhofer IPMS 09.2024 - 11.2024

- Automated an ultrasound measurement setup and programmed Universal Robots cobots in Python
- Developed a PyQt-based GUI for real-time synchronization and monitoring
- Implemented automated data logging and visualization pipelines using Python.



PROJECTS

Intelligent Machine Programming Lab

04.2026 - PRESENT

- Programmed robot manipulator tasks including drawing, erasing and task-board manipulation.
- Developed C++ and Python-based robot control modules for task execution and automation.
- Applied robot programming concepts to complete manipulation workflows from motion planning to execution.

Modeling and Control of Legged Robots

04.2026 - PRESENT

- Built a ROS2-based humanoid robot simulation workspace for the Talos platform.
- Integrated PyBullet, Pinocchio, TSID, and Drake/Pydrake for modeling, whole-body control, balance recovery, and walking generation.
- Implemented and analyzed modules for floating-base dynamics, ZMP/CMP/DCM balance strategies, LIP-based MPC, and footstep planning.

Hybrid Vision for Drone Detection

04.2026 - PRESENT

- Developed both a hybrid vision pipeline and event-based video reconstruction for drone detection by combining synchronized RGB and event-camera data.
- Implemented localization and classification output using bounding boxes and class labels.
- Built a web-based interface for data selection, video playback, and visualization of detection results.

Autonomous Drone Racing

10.2025 - 02.2026

- Developed an optimization-based autonomous drone racing pipeline for the Crazyflie nano quadrotor.
- Implemented spline-based reference generation with online replanning for gate traversal and obstacle avoidance.
- Designed an MPCC controller using acados and SQP-RTI, achieving stable sim-to-real deployment with real-world flight validation.

Computer Vision Challenge

06.2025 - 07.2025

- Built a MATLAB GUI for satellite image change detection and visualization.
- Implemented advanced visualization modes, including timelapse, difference highlighting, and heatmaps.
- Integrated image preprocessing workflows, including registration, cropping, and normalization, and validated results on unknown datasets.

Intelligent Upgrade of Quadruped Robot

04.2025 - 07.2025

- Extended the capabilities of a WAVEGO bionic dog-like robot by integrating a Raspberry Pi as the host controller.
- Integrated OpenAI API for interactive functionalities and decision-making support.
- Implemented YOLO-based object detection and tracking for basic autonomous navigation.